

1. Administrative & Study Identification

Full Study Title: A Study Evaluating the Efficacy and Safety of Giredestrant Compared With Physician's Choice of Endocrine Monotherapy in Participants With Previously Treated Estrogen Receptor-Positive, HER2-Negative Locally Advanced or Metastatic Breast Cancer (acelERA Breast Cancer) **Short Title/Acronym:** acelERA Breast Cancer **Protocol Number:** W042312 **NCT Number:** NCT04576455 **Sponsor/Company Name:** Roche (Hoffmann-La Roche / Genentech) **Investigational Product:** Giredestrant (GDC-9545; oral selective estrogen receptor degrader [SERD]) **Phase of Development:** Phase II **Medical Monitor:** Sponsor Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, specifically investigator-assessed Progression-Free Survival (INV-PFS), of giredestrant compared with physician's choice of endocrine monotherapy (PCET) in previously treated ER+, HER2-advanced breast cancer. **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Clinical Benefit Rate (CBR), Duration of Response (DoR), and safety/tolerability. An additional key secondary objective is to assess INV-PFS according to the presence of ESR1 mutations in circulating tumor DNA (ctDNA).

2.2 Background & Rationale Endocrine therapy (ET) is the mainstay of management for estrogen receptor-positive (ER+), HER2-negative advanced breast cancer. However, resistance often occurs, frequently driven by mutations in the ESR1 gene. Giredestrant is a highly potent, nonsteroidal, oral selective ER antagonist and degrader (SERD) that achieves robust ER occupancy. This study was conducted to evaluate if giredestrant offers superior clinical efficacy and an improved treatment experience compared to standard sequential endocrine monotherapies (like intramuscular fulvestrant or aromatase inhibitors), particularly following progression on prior lines that may have included CDK4/6 inhibitors.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Physician's Choice of Endocrine Therapy [PCET]: fulvestrant or an aromatase inhibitor) **Blinding:** Open-label **Randomization:** 1:1 ratio, stratified by visceral vs. nonvisceral disease, prior CDK4/6 inhibitor use, and prior

fulvestrant use. **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting)

3.2 Planned Enrollment & Duration Target N: 303 randomly assigned patients **Number of Sites:** 85 sites across 17 countries **Estimated Study Start:** 27-Nov-2020 **Estimated Primary Completion:** 18-Feb-2022 (Clinical Data Cutoff)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Post-/pre-/perimenopausal women, or men, age ≥ 18 years. For women who are premenopausal or perimenopausal and for men: willing to undergo and maintain treatment with approved LHRH agonist therapy. 2. Histologically confirmed diagnosis of ER-positive, HER2-negative locally advanced or metastatic adenocarcinoma of the breast, not amenable to treatment with curative intent. 3. Disease progression after 1-2 prior lines of systemic therapy for locally advanced/metastatic disease (must include ET for ≥ 6 months; ≤ 1 targeted agent allowed [e.g., CDK4/6i]; ≤ 1 chemotherapy regimen allowed; prior fulvestrant allowed if discontinued ≥ 28 days before random assignment). 4. Measurable disease per RECIST v1.1 or evaluable bone-only disease with ≥ 1 predominantly lytic bone lesion confirmed by CT or MRI. 5. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 6. Adequate organ function.

4.2 Exclusion Criteria 1. Prior treatment with an investigational or approved oral selective estrogen receptor degrader (SERD) (prior intramuscular fulvestrant was permitted if terminated at least 28 days prior to randomization). 2. Treatment with any investigational therapy within 28 days prior to randomization. 3. Advanced, symptomatic, visceral spread at risk of life-threatening complications. 4. Known active, uncontrolled, or symptomatic central nervous system (CNS) metastases, carcinomatous meningitis, or leptomeningeal disease. 5. Active cardiac disease or history of significant cardiac dysfunction. 6. Pregnant or breastfeeding.

5. Intervention & Treatment Plan

5.1 Investigational Regimen & Comparators Dosage/Frequency: Giredestrant (GDC-9545) 30 mg administered orally once daily. **Active Comparator (PCET):** Choice includes fulvestrant (Trade Name: Faslodex; ATC Code: L02BA03) or an aromatase inhibitor. **Route of Administration:** Oral (Giredestrant) / IM or Oral (PCET) **Duration of Treatment:** Administered in 28-day cycles continuously until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Luteinizing hormone-releasing hormone (LHRH) agonists for pre-/perimenopausal women and men (mandatory per local guidelines). **Prohibited:** Concurrent treatment with other investigational therapies, anticancer agents, or alternative SERDs. Dose reductions for giredestrant were not allowed.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
:---	:---	:---
:---	:---	Screening Day -28 to 0 Informed Consent, Medical History, Baseline Tumor Assessment (CT/MRI), ctDNA blood draw
	Baseline Day 0	Randomization, First Dose
	Every 8 Weeks	\$\pm\$ 7 Days Efficacy Assessment (RECIST v1.1 tumor imaging for the first 18 months, then Q12W), Safety Labs
	End of Study N/A	Treatment Discontinuation, Final Vital Signs, IP Accountability, Follow-up for Overall Survival

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Investigator-assessed Progression-Free Survival (INV-PFS). PFS was defined as the Kaplan-Meier estimate of time from randomization to the first occurrence of disease progression or death from any cause (whichever occurs first). Disease progression was defined as \$\ge\$ 20% increase in the sum of diameters of target lesions, unequivocal progression in non-target lesions, and/or appearance of new lesions. **Measurement Method:** Evaluated using computed tomography (CT) or magnetic resonance imaging (MRI) per RECIST v1.1 criteria.

7.2 Secondary & Exploratory Endpoints - Overall Survival (OS) - Objective Response Rate (ORR), as Determined by the Investigator According to RECIST v1.1 - Duration of Response (DoR), as Determined by the Investigator According to RECIST v1.1 - Clinical Benefit Rate (CBR) - INV-PFS assessed by ESR1 mutation status (evaluated via ctDNA testing)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection at every study visit; graded according to NCI CTCAE guidelines. **Serious Adverse Events (SAEs):** Standard reporting timelines applied (typically within 24 hours of investigator awareness). **Data Monitoring Committee (DMC):** Utilized an internal/external Independent Data Monitoring Committee to evaluate safety signals throughout the trial duration.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) / Full Analysis Set (FAS) for all efficacy endpoints; Safety Set for all patients receiving at least one dose of study drug. **Sample Size**

Justification: The targeted sample size of 303 was calculated to provide adequate statistical power to detect a clinically meaningful hazard ratio for INV-PFS between the giredestrant and active comparator arms. **Handling of Missing Data / Analysis:** Time-to-event endpoints (PFS, OS, DoR) were estimated using the Kaplan-Meier method. The 95% confidence interval for the median PFS was computed using the method of Brookmeyer and Crowley. Participants without the occurrence of disease progression or death as of the clinical cutoff were censored at the time of the last tumor assessment prior to the clinical cutoff. Differences between ORRs were analyzed using the stratified Cochran-Mantel-Haenszel test.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. Protocol approval obtained from independent ethics committees. **Monitoring Plan:** Standard sponsor-directed onsite and remote monitoring procedures to assure data integrity. **Audit Trail:** eCRF locking utilized; data vetted through secure, auditable clarification forms.

11. Study Identifiers & Governance

Primary ID: W042312 **Registry Number:** NCT04576455 **Posting ID:** 230276512594547872 **Sponsor/Lead Organization:** Roche **Study Title:** acELERA Breast Cancer **Official Regulatory Title:** A Phase II, Randomized, Open-Label, Multicenter Study Evaluating the Efficacy and Safety of GDC-9545 Compared With Physician's Choice of Endocrine Monotherapy in Patients With Previously Treated Estrogen Receptor-Positive, HER2-Negative Locally Advanced or Metastatic Breast Cancer

12. Clinical Framework (Breast Cancer Specific)

Disease Area: Estrogen Receptor-Positive, HER2-Negative, Locally Advanced or Metastatic Breast Cancer **Primary Condition / UMLS Definition:** Estrogen receptor positive and HER2 negative breast cancer (Preferred Name: Malignant neoplasm of breast) **Diagnosis Threshold:** Histologically confirmed ER+, HER2- disease progression after 1-2 lines of systemic therapy. **Medical Methodology:** Endocrine Monotherapy / Estrogen Receptor Degradation **Optimization Target:** Target suppression of ER

transcriptional activity to prolong PFS, including overcoming resistance via ESR1 mutations.

13. Study Procedures & Methodology

Management Pathway: Sequential Endocrine Monotherapy **Education Protocol (HCP):** Standard investigator meetings and protocol training for RECIST v1.1 and AE grading. **Education Protocol (Patient):** Thorough consent and study drug administration counseling (compliance with 28-day daily oral regimen). **Quality Audit Mechanism:** Independent central review validation checks and independent ethics committee audits.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until disease progression or death. **Onsite Visit Cadence:** Day 1 of every 28-day cycle for treatment and safety check. **Remote Monitoring Frequency:** Standard sponsor data sweeps every 4-8 weeks. **Checklist Review Interval:** Every 8 weeks for tumor imaging assessment (Q8W for 18 months, then Q12W).

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				